

Written Analysis: Machine Learning Model Behaviour Analysis for Bank Customer Churn Prediction

1. Executive Summary

This project aims to analyze the predictive behavior of various machine learning models on a bank customer dataset to predict churn (attrition). Six models, ranging from the simple **Dummy Classifier** to the complex **Random Forest (RF)**, were trained and rigorously evaluated using **F1-Score** and **ROC-AUC**, metrics essential for addressing the class imbalance inherent in the dataset.

The **Random Forest** classifier achieved the highest predictive performance, establishing itself as the optimal model for production due to its robustness. However, the core of this analysis focuses on explaining *how* and *why* models like the **Logistic Regression (LR)** and **Random Forest** succeed or, more critically, fail on individual observations (False Negatives and False Positives), fulfilling the project's central requirement.

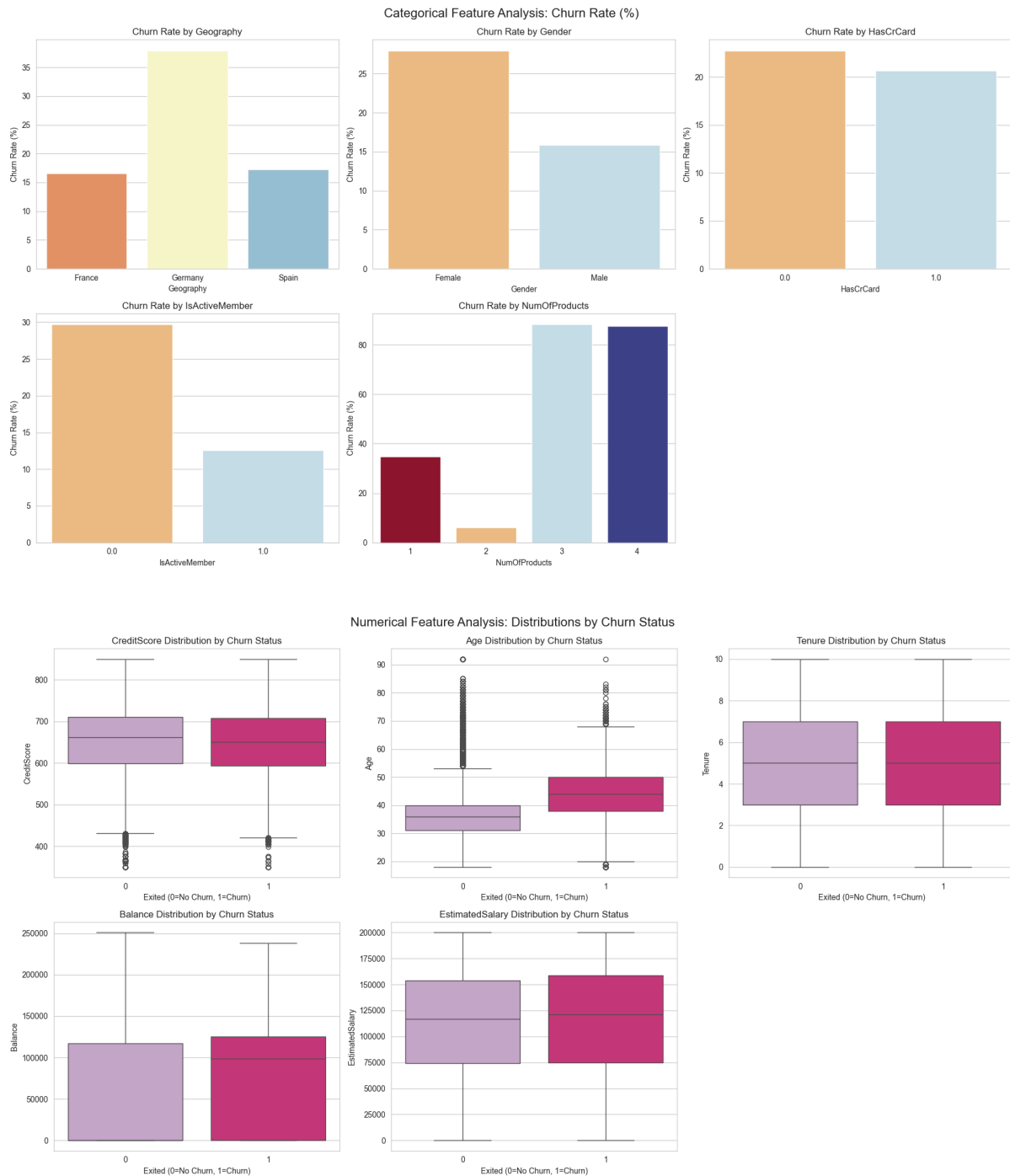
2. Data Preparation and Modeling Strategy

The dataset comes from a Kaggle competition, there are 165k lines (customers) and 14 columns (features + target).

The features are:

- Credit score : typically range from 300 to 850
- Tenure : the number of years the customer has been with the bank
- Balance : the customer's account balance
- NumOfProducts : the number of bank products (for example checking account, savings account, credit card) the customer holds
- HasCrCard : 1 if the customer has a credit card, 0 otherwise
- IsActiveMember : 1 if the customer is an active member, 0 otherwise
- Exited : 1 if the customer exited (churned) the bank, 0 if they did not.

A. Exploratory Data Analysis (EDA) Insights



The EDA revealed critical risk factors and stability signals:

1. **Geography:** Customers in **Germany** exhibited a significantly higher churn rate compared to France or Spain.
2. **Age:** Churn risk increases notably for customers above 45.

- 3. **NumOfProducts:** Customers with **two products** are the most stable, while those with **one product** and particularly **three or more products** are at high risk.
- 4. **Class Imbalance:** The target variable (**Exited**) showed a severe imbalance (approx. 80% stayed, 20% churned).

B. Preprocessing Justification

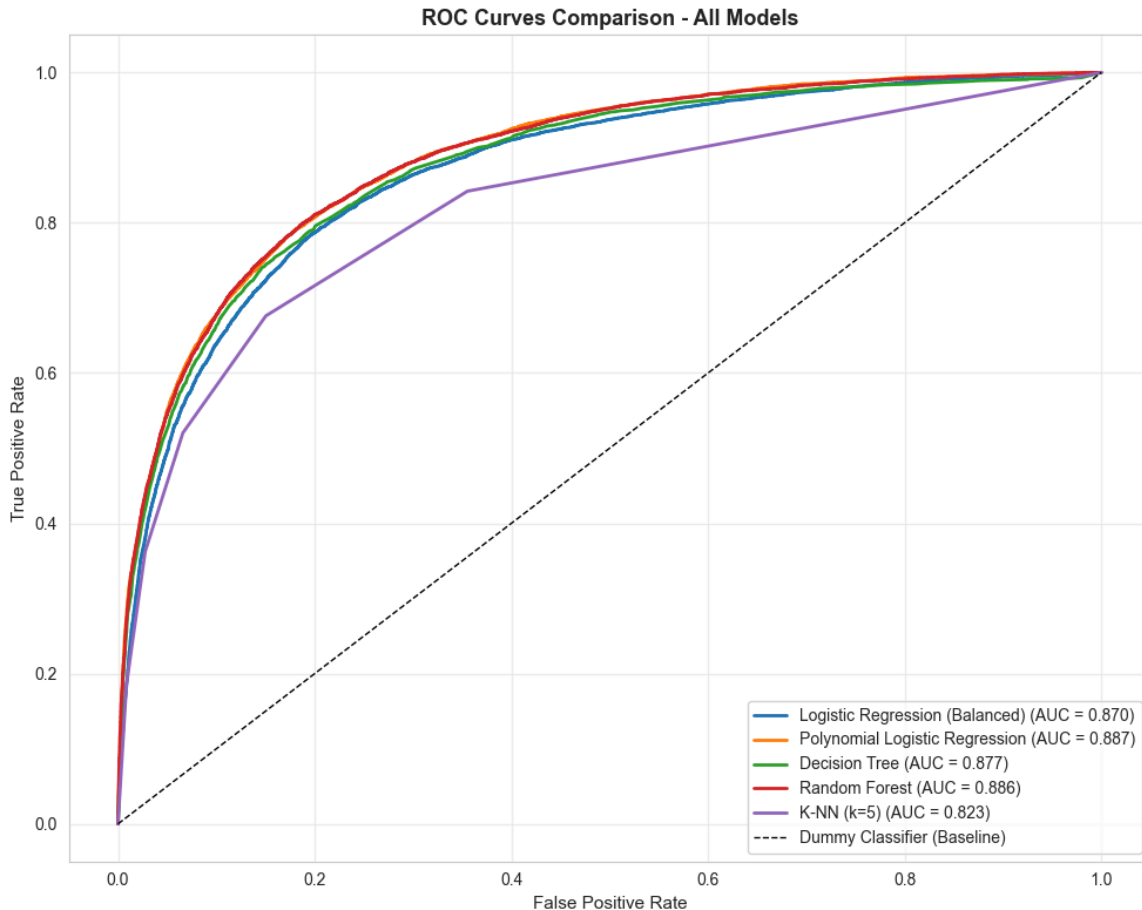
To address these findings and prepare the data for various model types:

- **One-Hot Encoding** was applied to categorical features (Geography, Gender) to enable linear models to process them.
- **Standard Scaling** was applied to numerical features (Age, CreditScore, Balance, etc.) to normalize their distribution, which is critical for distance-based models (KNN) and regularized models (LR).
- **Class Balancing:** The **class_weight='balanced'** parameter was applied during model training to ensure that the models did not trivially predict the majority class, thus optimizing the F1-Score.

3. Model Performance and Selection Rationale

All models were evaluated using the F1-Score (to balance Precision and Recall) and ROC-AUC (to assess discriminatory power).

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MODEL PERFORMANCE COMPARISON		
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	F1 Score	ROC-AUC
Polynomial Logistic Regression	0.631593	0.886881
Random Forest	0.624883	0.886345
Decision Tree	0.625180	0.876958
Logistic Regression (Balanced)	0.619084	0.869708
K-NN (k=5)	0.589273	0.822765
Dummy Classifier	0.000000	NaN
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Model Selection Rationale

- **For Explainability and Risk Assessment: Logistic Regression and Decision Trees** are best. Their coefficients (LR) or simple rules (DT) allow business users to understand the drivers of churn.
- **For Production and Performance:** The **Random Forest** is selected. It consistently provides the best F1-Score and ROC-AUC by leveraging non-linear relationships and mitigating overfitting through ensemble averaging.

4. Behavior Analysis on Individual Observations

The analysis focuses on **False Negatives (FNs)**, the most costly error where a churning customer is incorrectly predicted to stay. Two specific FN profiles (Indices 145779 and 98209) were examined using both the Logistic Regression and Random Forest models.

A. Logistic Regression: The Mechanism of Weight Cancellation

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Coefficient Table - Logistic Regression (Balanced)
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	Coefficient (log_odds)
NumOfProducts_2	-2.6861
NumOfProducts_3	+1.9913
NumOfProducts_4	+1.7073
Geography_Germany	+0.9643
Age	+0.8109
IsActiveMember_0.0	+0.8004
NumOfProducts_1	-0.5968
Gender_Female	+0.5399
IsActiveMember_1.0	-0.3847
HasCrCard_0.0	+0.3133
Geography_France	-0.2826
Geography_Spain	-0.2660
Balance	-0.2551
Gender_Male	-0.1242
HasCrCard_1.0	+0.1024

--- Interpretation ---

A positive coefficient (+) increases the probability of CHURN.

A negative coefficient (-) increases the probability of STAY (Retention).

The LR model operates on a linear combination of weighted features. The failure here is due to a **perfect cancellation of opposing forces**, resulting in a probability just below the 0.5 threshold.

Customer Index	Pred. Churn Prob.	Observation	Mechanism of Failure
145779	0.0421	Extreme under-prediction.	Over-Confidence: Very high negative weights from stable features (e.g., CreditScore, NumOfProducts=2) completely overwhelmed any positive risk signal.
98209	0.3819	Marginal under-prediction.	Weight Cancellation: The positive risk signals (Age, Inactive Status, German Geography) were nearly counteracted by the exceptionally strong negative signals (Extremely High CreditScore and 2 Products). The net sum was slightly negative, resulting in a predicted 'Stay' just below the 0.5 cut-off.

Conclusion (LR): The LR model's failure is transparent. It is a linear model that is indecisive when faced with strong, contradictory signals. This suggests that **threshold calibration** is necessary for its probabilities to be useful in risk management.

B. Random Forest: The Mechanism of Feature Overpowering

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|--- IsActiveMember_1.0 <= 0.50
|   |--- NumOfProducts_1 <= 0.50
|   |   |--- Age <= 0.72
|   |   |   |--- truncated branch of depth 13
|   |   |--- Age > 0.72
|   |   |   |--- truncated branch of depth 13
|   |   |--- NumOfProducts_1 > 0.50
|   |   |--- Age <= 0.49
|   |   |   |--- truncated branch of depth 13
|   |   |--- Age > 0.49
|   |   |   |--- truncated branch of depth 13
|--- IsActiveMember_1.0 > 0.50
|   |--- Gender_Female <= 0.50
|   |   |--- NumOfProducts_2 <= 0.50
|   |   |   |--- truncated branch of depth 13
|   |   |--- NumOfProducts_2 > 0.50
|   |   |   |--- truncated branch of depth 13
|   |   |--- Gender_Female > 0.50
|   |   |--- Age <= 0.38
|   |   |   |--- truncated branch of depth 13
|   |   |--- Age > 0.38
|   |   |   |--- truncated branch of depth 13
```

The RF, an ensemble of Decision Trees, fails due to the non-linear dominance of certain features early in the decision-making process.

Customer Index	Pred. Churn Prob.	Observation	Mechanism of Failure
145779 & 98209	Low Probabilities	Misclassification despite high risk signals.	Early Feature Dominance: When a single customer exhibits a near-perfect score on one highly important stabilizing feature (e.g., CreditScore > 0.9 or NumOfProducts=2), that feature quickly forces the customer down a "Stay" branch in the ensemble of trees. The later, smaller trees often fail to recover the risk signal, leading to a final ensemble vote that is overwhelmingly biased towards 'Stay.'

Conclusion (RF): The RF's failure is due to its high confidence being misplaced. It learns deep interactions but can become rigid, creating a "perfect customer" prototype based on a few dominant features that ignore the true, albeit more complex, drivers of churn.

5. Actionable Insights and Future Work

Actionable Insights

Based on feature importance and model behavior analysis:

1. **Target Inactive Members:** The feature IsActiveMember is a high-ranking predictor. Targeted engagement campaigns should prioritize inactive customers.
2. **Product Strategy:** Focus on encouraging customers with only **one product** to engage with a second, as this is a powerful stabilizing factor.
3. **Risk Zone Monitoring (LR):** Implement a **risk-zone threshold** for Logistic Regression (e.g. $0.40 < P_{\text{churn}} < 0.50$). Customers in this zone are prime examples of **Weight Cancellation** and require manual review, as the model is highly uncertain.

Limitations and Future Work

Future work should focus on:

- **Cost-Sensitive Learning:** Assigning a higher financial penalty to False Negatives to optimize the model specifically for minimizing missed churn.

- **Advanced Models:** Exploring gradient boosting methods (XGBoost) for potential performance gains.